LE CHATELIER'S PRINCIPLE

OBJECTIVES

- To demonstrate/observe Le Chatelier's Principle by examining several chemical systems.

INTRODUCTION

Le Chatelier's Principle states:
If a chemical reaction at equilibrium is subjected to a change in conditions that displaces it from equilibrium, then the reaction adjusts toward a new equilibrium state. The reaction proceeds in the direction that - at least partially - offsets the change in conditions.

A change in conditions that will affect the reaction equilibrium may include the concentration of a reactant or product, reaction volume or applied pressure, or temperature. During this lab you will examine several chemical systems, each in a state of dynamic equilibrium, and then observe the physical changes that occur when a change of conditions, or stress, is applied to each system.

An example of Le Chatelier's Principle can be found in the endothermic reaction:

$$N_2O_{4\,(g)} \leftrightarrow 2\ NO_{2\,(g)}$$



The equilibrium position can be shifted by changing the temperature. When heat is added and the temperature increases, the reaction shifts forward and the flask turns reddish brown due to an increase in NO_2. When heat is removed and the temperature decreases, the reaction shifts to the reverse direction, and the flask turns colorless due to an increase in N_2O_4. The reverse reaction is exothermic.

Figure 1: The reaction of $N_2O_{4\,(g)}$ (colorless) $\leftrightarrow$ $2\ NO_{2\,(g)}$ (red).

You will be investigating the following chemical systems:

1. Chromate – dichromate equilibrium in aqueous solution.

$$CrO_{4\,(aq)}^{2-} + H_{(aq)}^{+} \leftrightarrow Cr_2O_{7\,(aq)}^{2-} + H_2O_{(l)} \qquad \text{(unbalanced)}$$

2. Acetic Acid – Acetate ion equilibrium in aqueous solution.

$$CH_3COOH_{(aq)} \leftrightarrow CH_3COO_{(aq)}^{-} + H_{(aq)}^{+}$$

3. Tetrachlorocobaltate(II)-hexaquacobalt(II) equilibrium in aqueous solution.

$$CoCl_{4\,(aq)}^{2-} + 6\ H_2O_{(l)} \leftrightarrow Co(H_2O)_{6\,(aq)}^{2+} + 4\ Cl_{(aq)}^{-}$$

4. Solid ammonium chloride in equilibrium with aqueous ammonium ions and chloride ions.

$$NH_4Cl_{(s)} + H_2O_{(l)} \leftrightarrow NH_{4\,(aq)}^{+} + Cl_{(aq)}^{-}$$

PRE-LAB Exercise: You will be using many ionic compounds in this lab. Write the balanced net ionic equations for the following ionic compounds dissolved in water:

K_2CrO_4, H_2SO_4, $NaOH$, CH_3COOH, $NaCH_3COO$, HCl, and NH_4Cl.

An example balanced net ionic equation for $K_2Cr_2O_7$ is provided below to help guide your pre-lab exercise:

$$K_2Cr_2O_{7\,(s)} \leftrightarrow 2\ K_{(aq)}^{+} + Cr_2O_{7\,(aq)}^{2-}$$

In-lab: Investigate these reactions then answer the questions in your lab notebook. Make sure to record all observations in your notebook as well as the answers to the questions asked. **There is no need for a separate conclusion section this week. The answers to these questions will serve as your conclusion for this week.**

PROCEDURE

PART A. Chromate – dichromate equilibrium in aqueous solution.

$$CrO_{4\,(aq)}^{2-} + H_{(aq)}^{+} \leftrightarrow Cr_2O_{7\,(aq)}^{2-} + H_2O_{(l)}$$

(chromate ion, CrO_4^{2-}, yellow) (dichromate ion, $Cr_2O_7^{2-}$, orange)

1. Place on dropperful (approx. 1 mL) of a 0.1 M K_2CrO_4 (potassium chromate) solution into a small, clean, dry test tube. Observe the color.

2. Add 3.0 M H_2SO_4 (sulfuric acid) one drop at a time, while stirring with a glass rod, until you observe a significant color change.

3. Next, add 1.0 M $NaOH$ (sodium hydroxide) one drop at a time, while stirring with a glass rod, until you observe another significant color change.

4. Discard the solution into the waste container and wash your test tube and stir rod with water.

QUESTIONS PART A:

1. Balance the chromate/dichromate equilibrium equation:

$$CrO_{4\,(aq)}^{2-} + H_{(aq)}^{+} \leftrightarrow Cr_2O_{7\,(aq)}^{2-} + H_2O_{(l)} \quad \text{(unbalanced)}$$

2. What was the color of the K_2CrO_4 solution that you started with? If the above equilibrium is established in the original solution, which species, $CrO_{4\,(aq)}^{2-}$ or $Cr_2O_{7\,(aq)}^{2-}$ appears to be present in the greatest amount? (Hint: look at the color).

3. What color did you observe after adding the 3.0 M H_2SO_4 solution?

4. What color did you observe after adding the 1.0 M $NaOH$ solution?

5. Based on Le Chatelier's Principle, when the sulfuric acid is added:

    a. Which reaction was favored, the forward or reverse reaction? What evidence supports your answer?

    b. How does this affect the amount of $Cr_2O_7^{2-}$ions present? How does this affect the amount of CrO_4^{2-}ions present?

6. Based on Le Chatelier's Principle, when sodium hydroxide is added:

a. Which reaction was favored, the forward or reverse reaction? What evidence supports your answer? Why would this reaction be favored? (Hint: what does OH^- do with H^+?)

b. How does this affect the amount of $Cr_2O_7^{2-}$ ions present? How does this affect the amount of CrO_4^{2-} ions present?

PART B. Acetic Acid – Acetate ion equilibrium in aqueous solution.

$$CH_3COOH_{(aq)} \leftrightarrow CH_3COO^-_{(aq)} + H^+_{(aq)}$$

(acetic acid, colorless) (acetate ion, colorless) (hydrogen ion, colorless)

NOTE: Below is the structure methyl orange indicator in acidic and basic environments. In solutions that are **more acidic, methyl orange displays a visible red color.** In solutions that are **less acidic, methyl orange appears yellow.**

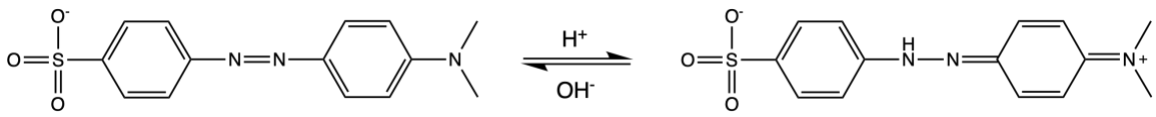


Figure 2. In an alkaline solution, methyl orange is in the yellow form (left). When an acid is added, the hydrogen ion bonds to the nitrogen atom to give the red form (right).

1. Place one dropperful (approx.1 mL) of a 1.0 M CH_3COOH solution into a small, clean, dry test tube. Add one drop of methyl orange indicator. Observe the color.

2. Use a scoopula to add a small portion of solid sodium acetate, $NaCH_3COO$, (just the scoopula tip is sufficient) to the solution. Stir the solution with a glass stir rod until the solid dissolves. Keep adding small portions of sodium acetate while stirring until you observe a color change.

3. Discard the solution into the waste container and rinse the test tube and stir rod with water.

QUESTIONS PART B:

1. What was the color of the solution after adding the drop of methyl orange?

2. What color did you observe after adding the sodium acetate?

3. Based on Le Chatelier's Principle, when sodium acetate was added:

    a. Which reaction, the forward or reverse was favored? What evidence supports this? Why would this reaction be favored?

    b. How does this affect the amount of H^+ ions present? How does this affect the amount of CH_3COOH molecules present?

PART C. Tetrachlorocobaltate(II)-hexaquacobalt(II) equilibrium in aqueous solution.

$$CoCl_{4\,(aq)}^{2-} + 6\ H_2O_{(l)} \leftrightarrow Co(H_2O)_{6\,(aq)}^{2+} + 4\ Cl_{(aq)}^{-}$$

(tetrachlorocobaltate(II) ion, blue) (hexaquacobalt(II) ion, pink)

1. Place 5-10 drops of 0.1 M $CoCl_2$ (cobalt chloride) solution into a small, clean, dry test tube. Observe the color. What species does $CoCl_2$ in water form based on the color?

2. Add concentrated HCl one drop at a time, while stirring with a glass stir rod until you observe a color change.

3. Add distilled water one drop at a time while stirring until you see a color change.

4. Discard the solution into the waste beaker and rinse with water.

QUESTIONS PART C:

1. What was the original color of the cobalt chloride solution?

2. What color did you observe after adding the concentrated HCl?

3. What color did you observe after adding the distilled water?

4. Based on Le Chatelier's Principle, when HCl is added:

    a. Which reaction, the forward or reverse was favored? What evidence supports this? Why would this reaction be favored?

    b. How does this affect the amount of $Co(H_2O)_6^{2+}$ ions present? How does this affect the amount of $CoCl_4^{2-}$ ions present?

PART D. Solid ammonium chloride in equilibrium with aqueous ammonium and chloride ions.

$$NH_4Cl_{(s)} + H_2O_{(l)} \leftrightarrow NH_{4\,(aq)}^{+} + Cl_{(aq)}^{-}$$

(ammonium chloride) (ammonium ion) (chloride ion)

1. Pour roughly 75 mL of water into a 150 mL beaker and heat to boiling on a hot plate.

2. While waiting for the water to boil, place one dropperful (1 mL) of a saturated NH_4Cl solution into a small, clean, dry test tube.

3. Add concentrated HCl one drop at a time, while stirring with a glass rod until you observe a change.

4. Using a test tube clamp, place the test tube into the boiling water bath. Leave it in the bath for a few minutes, stirring the contents until you observe a change.

5. Use the scoopula tip to add a few crystals of solid ammonium chloride to the bottom of a **NEW** clean, dry test tube. Touch the bottom of the test tube to quickly assess the temperature. Now add 10 drops of distilled water and feel the bottom of the tube again, is there any temperature change?

6. Discard the solution into the waste beaker and clean out all glassware and return to the cart.

QUESTIONS PART D:

1. Based on Le Chatelier's Principle, when HCl is added:

    a. What did you observe?

    b. Which reaction, the forward or reverse was favored? What evidence supports this?

    c. What did you observe when you placed this test tube in the boiling water bath?

    d. Based on the observation in part c, which reaction is endothermic and which is exothermic?

2. Based on Le Chatelier's Principle, when H_2O is added to solid ammonium chloride:

    a. What temperature change, if any, did you feel?

    b. Which reaction, the forward or reverse was favored? What evidence supports this?

    c. What does this temperature change indicate, and is it consistent with your findings in question 1?

Make sure to record all observations in your notebook as well as the answers to the questions asked. There is no need for a separate conclusion section this week. The answers to these questions will serve as your conclusion for this week.